

KARAMAY, China

WMO: 512430

Lat: 45.6101N

Lon: 84.8454E

Elev: 457

StdP: 95.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-23.1	-29.0	0.3	-25.3	-26.1	0.4	-22.5	9.2	-7.7	4.7	-9.7	0.8	340	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	36.6	18.2	35.1	17.8	33.7	17.5	19.7	31.1	19.1	31.1	18.5	30.6	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.2	23.9	14.8	11.1	24.3	13.6	10.2	24.6	58.5	30.9	56.2	31.1	54.3	30.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.2	7.2	-26.1	39.4	2.8	1.8	-28.1	40.7	-29.8	41.8	-31.4	42.8	-33.5	44.1		
(5)			-26.3	21.0	2.8	1.0	-28.3	21.7	-30.0	22.3	-31.5	22.8	-33.6	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-15.8	-10.8	1.8	14.0	20.5	26.1	27.7	26.1	19.8	11.0	0.5	-11.0	(6)
(7)		DBStd	15.91	5.03	5.26	7.12	5.19	4.46	3.69	3.26	3.54	4.53	4.83	5.76	5.22	(7)
(8)		HDD10.0	2661	801	583	266	23	1	0	0	0	1	48	286	652	(8)
(9)		HDD18.3	4273	1059	816	511	145	29	1	0	1	35	232	534	910	(9)
(10)		CDD10.0	2390	0	0	13	142	327	483	550	500	296	78	1	0	(10)
(11)		CDD18.3	961	0	0	0	14	97	233	292	242	80	3	0	0	(11)
(12)		CDH23.3	10406	0	0	0	91	832	2607	3529	2712	625	9	0	0	(12)
(13)		CDH26.7	4753	0	0	0	18	286	1216	1775	1259	198	1	0	0	(13)
(14)	Wind	WSAvg	2.6	1.0	1.3	2.5	3.7	3.8	3.7	3.3	3.2	3.0	2.7	2.0	1.2	(14)
(15)	Precipitation	PrecAvg	104	3	2	4	7	14	15	22	16	7	6	5	5	(15)
(16)		PrecMax	227	12	10	16	24	51	56	70	41	26	18	24	18	(16)
(17)		PrecMin	58	0	0	0	0	1	0	2	1	0	0	0	0	(17)
(18)		PrecStd	39	3	2	4	6	12	14	15	10	6	5	6	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	2.3	20.2	29.1	34.5	37.4	39.2	38.0	34.0	24.6	14.2	1.4	(19)
(20)			MCWB	-3.8	-1.1	9.6	13.4	16.1	18.1	19.4	18.5	16.4	12.4	7.8	-1.1	(20)
(21)		2%	DB	-5.3	-0.1	16.7	25.7	31.8	35.7	37.0	35.8	31.0	21.7	11.3	-0.5	(21)
(22)			MCWB	-6.4	-2.4	8.4	12.1	15.3	17.4	18.8	18.0	15.5	11.1	6.0	-2.2	(22)
(23)		5%	DB	-7.4	-2.1	14.4	23.7	29.8	34.1	35.3	34.2	28.9	19.7	9.3	-2.3	(23)
(24)			MCWB	-8.3	-3.9	7.4	11.2	14.6	17.1	18.3	17.7	14.7	10.4	4.7	-3.6	(24)
(25)		10%	DB	-9.0	-3.8	12.1	21.6	27.8	32.5	33.8	32.5	26.9	17.8	7.7	-3.9	(25)
(26)			MCWB	-9.7	-5.1	6.2	10.5	13.9	16.9	18.0	17.3	13.8	9.5	3.6	-5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	-0.4	10.6	14.6	17.2	20.0	21.2	20.6	17.2	13.6	8.5	-0.3	(27)
(28)			MCDWB	-2.4	1.5	18.7	25.0	30.6	30.7	34.9	29.9	29.8	21.8	12.9	0.7	(28)
(29)		2%	WB	-6.4	-2.3	8.9	13.4	16.2	19.0	20.0	19.4	16.3	12.1	6.6	-2.1	(29)
(30)			MCDWB	-5.4	-0.5	15.8	22.6	28.6	31.5	31.3	31.3	28.2	20.0	10.6	-0.8	(30)
(31)		5%	WB	-8.2	-3.8	7.6	12.3	15.5	18.3	19.3	18.6	15.5	11.0	5.1	-3.4	(31)
(32)			MCDWB	-7.5	-2.1	14.1	21.7	27.2	30.8	30.9	31.0	26.8	18.4	9.2	-2.3	(32)
(33)		10%	WB	-9.7	-5.1	6.2	11.3	14.7	17.7	18.8	18.0	14.6	10.0	3.7	-5.1	(33)
(34)			MCDWB	-9.0	-3.8	11.8	20.4	26.0	30.1	30.6	30.4	25.4	17.3	7.3	-3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.5	6.1	7.4	9.1	9.3	9.0	8.9	9.3	8.8	7.6	5.4	4.7	(35)
(36)			MCDBR	5.7	6.5	9.5	10.8	10.8	10.2	10.1	10.7	10.5	9.4	7.1	4.9	(36)
(37)			MCWBR	4.9	4.9	5.1	4.9	4.3	3.6	3.5	3.8	4.6	4.8	4.6	3.7	(37)
(38)		5% WB	MCDWR	5.5	6.0	9.0	9.3	9.6	9.1	9.0	9.4	8.5	6.9	4.5	4.5	(38)
(39)			MCWBR	4.7	4.7	5.0	4.6	4.0	3.3	3.3	3.5	4.4	4.5	4.5	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.397	0.443	0.483	0.458	0.466	0.461	0.439	0.419	0.414	0.387	0.363	(40)	
(41)		taud	1.994	1.959	1.932	1.901	2.022	2.069	2.125	2.159	2.139	2.058	2.058	2.016	(41)	
(42)		Ebn at Noon	677	739	766	776	814	808	807	809	788	715	651	629	(42)	
(43)		Edh at Noon	109	137	162	183	167	161	150	140	131	122	102	96	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.79	2.75	4.07	5.42	6.58	6.74	6.54	5.95	4.79	3.11	1.84	1.35	(44)	
(45)		RadStd	0.10	0.12	0.34	0.29	0.30	0.28	0.36	0.24	0.24	0.21	0.17	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-0.46	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page